BRIEF REPORT

Mitochondrial Donation in a Reproductive Care Pathway for mtDNA Disease

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SUMMARY

Pathogenic variants in mitochondrial DNA (mtDNA) are a common cause of severe, often fatal, inherited metabolic disease. A reproductive care pathway was implemented to provide women carrying pathogenic mtDNA variants with reproductive options. A total of 22 women with pathogenic mtDNA variants have commenced or completed pronuclear transfer (and thus receipt of a mitochondrial donation), and there have been 8 live births. All 8 children were healthy at birth, with no or low levels of mtDNA heteroplasmy in blood. Hyperlipidemia and cardiac arrhythmia developed in a child whose mother had hyperlipidemia during pregnancy; both of the child's conditions responded to treatment. Infant myoclonic epilepsy developed in another child, with spontaneous remission. At the time of this report, all the children have made normal developmental progress. (Funded by the U.K. National Health Service and others.)

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Pathogenic variants in mitochondrial DNA (mtDNA) are a common cause of severe inherited metabolic disease.[1,2] Mitochondrial DNA is present in multiple copies within each cell. For many pathogenic variants, the relative abundance (heteroplasmy) of mutated copies is a key factor in determining the severity of disease, with higher levels of heteroplasmy causing disease in a diverse range of organs.[3] However, other pathogenic mtDNA variants are typically homoplasmic (all copies are mutated) and exhibit a characteristic pattern of isolated organ involvement with variable penetrance.[4,5]

Despite the prevalence and severity of mtDNA disease, treatment options are limited.[6] The exclusive maternal transmission of mtDNA means that children of women with homoplasmic pathogenic mtDNA variants are obligate homoplasmic carriers. The outcome for children born to mothers with heteroplasmy is less predictable; during female germ-cell development, a mitochondrial genetic bottleneck leads to a random shift in heteroplasmy, which results in offspring with variable mutant loads.[7,8]

For women considering treatment through assisted reproductive technology to lower the risk of serious mtDNA-related disease in their offspring, options include preimplantation genetic testing (PGT).[9,10] However, for women with high levels of mtDNA heteroplasmy or homoplasmic pathogenic mtDNA variants, PGT is not an option and mitochondrial donation has been considered as a possibility.[11]

After careful preclinical testing of mitochondrial donation with the use of pro-

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nuclear transfer[12] and a change in the Human Fertilisation and Embryology Act (United Kingdom), a new comprehensive National Health Service (NHS) care pathway[13] was established to provide informed reproductive choices for women; one of these choices is to accept mitochondrial donation through pronuclear transfer. Here, we describe this pathway and present some data on the genetic and clinical outcomes of pregnancies and children born after mitochondrial donation. In an accompanying article in this issue of the *Journal*, Hyslop et al.[14] describe the corresponding embryology data.

METHODS

The mitochondrial reproductive care pathway that was implemented in 2017 (the NHS Mitochondrial Reproductive Care Pathway) is available to all women living in the United Kingdom who harbor pathogenic mtDNA variants, with an underlying principle to provide these women with an informed reproductive choice that can be implemented in a regulated environment.[13,15] The pathway comprises the Mitochondrial Reproductive Advice Clinic (MRA-C) and the Mitochondrial Assisted Reproductive Technology Clinic (MART-C) where women and their partners undergo detailed clinical review (see the Methods section in the Supplementary Appendix, available with the full text of this article at NEJM.org).

As part of the licensing conditions, women who proceed with mitochondrial donation are offered close monitoring of the pregnancy and follow-up of the offspring (Fig. S1 in the Supplementary Appendix). Throughout the process, any health concerns are discussed and, when indicated, assessed by the NHS Highly Specialised Mitochondrial Reproductive Care Pathway team, including assessment of the long-term health consequences for all children born.

To obtain funding for mitochondrial donation, we devised and registered a clinical study to assess the children at 18 months of age by means of the Bayley Scales of Infant and Toddler Development, 3rd edition[16] (National Institute of Health and Care Research [NIHR] Central Portfolio Management System identification number, 31755; ClinicalTrials.gov number, NCT04113447). At the time that the manuscript was submitted, only two children had completed the 18-month assessment; therefore, we are not reporting the results of the clinical trial but the efficacy and safety of mitochondrial donation before that point.

RESULTS

REPRODUCTIVE CARE PATHWAY

A total of 196 referrals, which broadly reflected the demographic characteristics of the U.K. population (Table S1), were reviewed by the Referral Assessment Multidisciplinary Team. A total of 4 referrals were not considered further owing to insufficient referral information or an incorrect diagnosis (Fig. 1). Before the offer of an appointment in the MRA-C, 23 women withdrew from the process, of whom 9 were already pregnant. A total of 6 women have appointments pending.

Of the 163 women evaluated in the MRA-C (not including the 6 women in whom evaluation is pending), 133 have received or are receiving care in the MART-C, 2 of whom are participating in healthy-lifestyle programs to decrease their body-mass index (Fig. 1). A total of 30 women were unable or opted not to proceed further; most of these women viewed the process as "fact finding" to inform their future reproductive decision making, and 3 had conceived naturally before their planned consultation in the MART-C.

After MART-C assessment of their reproductive options, 51 women did not proceed in the program. Reasons included limitations to undergoing in vitro fertilization, such as a very low ovarian reserve (6 women) and age (1 woman), or risks involved in pregnancy owing to poor maternal health. Many women declined at this stage because of personal priorities and timing, often with a desire to proceed in the future. A total of 5 women were offered a reproductive intervention but declined without providing a reason.

Of the 70 women who proceeded with a reproductive option, 36 (in whom 10 different pathogenic mtDNA variants were represented) were offered PGT. The use of PGT resulted in the birth of 13 children (Fig. 1) since the start of the care pathway.

MITOCHONDRIAL DONATION

Applications for Mitochondrial Donation

All 32 applications that were submitted to the Human Fertilisation and Embryology Authority for mitochondrial donation with the use of pro-

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nuclear transfer were approved; 4 applications are pending (Fig. 1). Ten pathogenic mtDNA variants (all in Cambridge reference sequence number NC_012920.1) were represented across the 32 women: m.11778G→A (in 9 women), m.3243A→G (in 7), m.3460G→A (in 4), m.8344A→G (in 4), m.4300A→G (in 3), m.3250T→C (in 1), m.3260A→G (in 1), m.3635G→A (in 1), m.7510T→C (in 1), and m.14484T→C (in 1). Of these 32 women, 3 had previously had embryos genetically tested (1 of them, before the introduction of the care pathway) but did not have any transferred owing to high levels of heteroplasmy in each tested embryo.

Pregnancies after Mitochondrial Donation

Ultrasonography at 7 weeks confirmed eight pregnancies, from which there have been eight live births (four female and four male), including a set of monozygotic twins, and one ongoing pregnancy (Table 1). There have been no miscarriages after ultrasonographic confirmation of pregnancy at 7 weeks' gestation. Six pregnancies were without complication. In one woman who was homoplasmic for the m.4300A→G variant and had a preexisting hypertrophic cardiomyopathy, paroxysmal atrial fibrillation developed during preparation for embryo transfer, which led to treatment with a beta-blocker and thromboprophylaxis. This patient also had a clinically significant complication of severe hypertriglyceridemia during pregnancy (serum triglyceride level, 77.17 mmol per liter [6835 mg per deciliter; normal range, 0.55 to 1.69 mmol per liter {49 to 150 mg per deciliter}]; serum total cholesterol level, 23.2 mmol per liter [897 mg per deciliter; normal value, <5.2 mmol per liter {200 mg per deciliter}]), which improved with restriction of dietary fat to less than 20 g per day and resolved within a week post partum.

FOLLOW-UP OF CHILDREN

Assessment of mtDNA Variant Levels

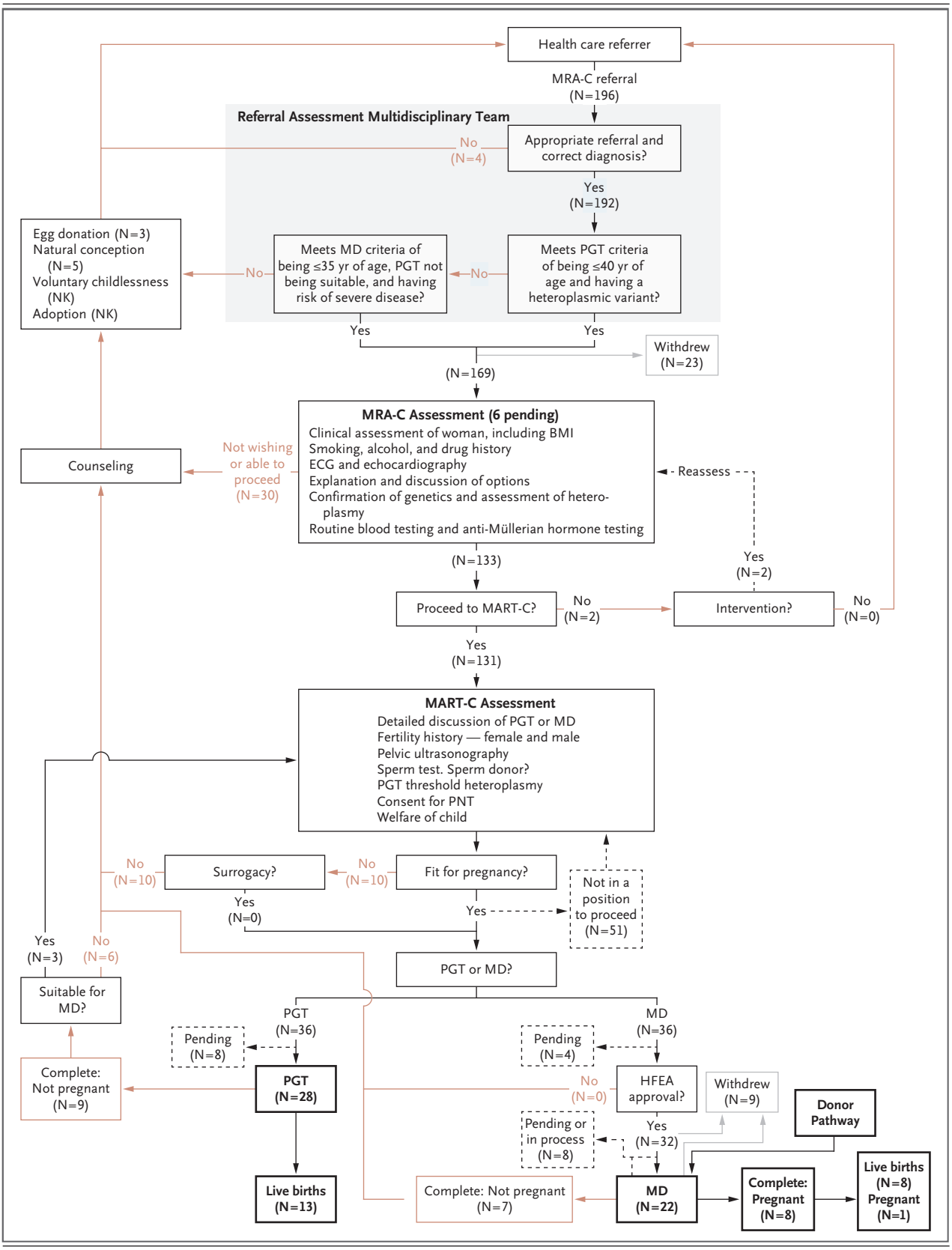
All the babies were born between 36 weeks 1 day and 42 weeks 2 days by normal vaginal delivery or elective cesarean section for potential maternal-health concerns or, in one case, placenta previa. Newborn and placental weights and Apgar scores were within normal ranges (Table 2). The level of heteroplasmy, available in each of the eight children, was below the threshold for clinical disease. Five children had undetectable levels at birth. A child whose mother was homoplasmic for the m.4300A→G variant had levels of 5% and 9% in blood and urine, respectively, at birth and undetectable levels in blood at 18 months. A child (born to a mother with the m.3260A→G variant) had 16% and 20% heteroplasmy in blood and urine, respectively, at birth, and another child (born to a mother with the m.11778G→A variant) had 12% and 13% heteroplasmy in blood and urine, respectively, at birth.

Assessment of Children

All eight neonates were healthy at birth and appear to be making normal developmental progress (see the Results section in the Supplementary Appendix). At the time of this report, three children are between 0 and 5 months of age, two are between 6 and 11 months of age, one is between 12 and 17 months of age, one is between 18 and 23 months of age, and one is at least 24 months of age. To date, five children have had no reported medical problems. An infant born to a mother with the m.3460G→A variant, who had no detectable maternal mtDNA at birth, had brief startles (head and neck flexion with eye blink) from 7 months of age. Neurologic, developmental, and general examinations were normal, as were routine blood tests and an interictal electroencephalogram. The child received a diagnosis of myoclonic epilepsy of infancy, a rare, typically self-limiting condition that occurs in otherwise healthy and developmentally normal children.[17] After 3 months, myoclonic jerks ceased without treatment, and developmental progress remains normal in all domains. Another child had a urinary tract infection that responded promptly to antibiotic treatment.

One breast-fed infant, born to the woman with the m.4300A→G variant who had gestational severe hypertriglyceridemia, had prolonged jaundice with mildly elevated levels of liver enzymes (elevated alanine aminotransferase and γ-glutamyltransferase levels with conjugated hyperbilirubinemia), an enlarged hyperechogenic liver (indicative of hepatic steatosis) on ultrasonography, and hyperlipidemia (serum triglyceride level, 7.5 mmol per liter [664 mg per deciliter; normal value, <1.15 mmol per liter {102 mg per deciliter}]). Dietary restriction of fat intake led to resolution of the hyperlipidemia and fatty liver over the course of 3 months. An echocardiogram, obtained at approximately 4 months of age, revealed a dilated

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Figure 1 (facing page). Reproductive Care Pathway for Women with Pathogenic mtDNA Variants.

Women are referred for reproductive advice and potential assisted reproductive technology to limit the transmission of the pathogenic variant of mitochondrial DNA (mtDNA). The numbers of women referred and progressing through each part of this pathway are shown. Women are assessed at the Mitochondrial Reproductive Advice Clinic (MRA-C) or Mitochondrial Assisted Reproductive Technology Clinic (MART-C) before either preimplantation genetic testing (PGT) or pronuclear transfer (PNT). The 36 women who were offered mitochondrial donation (MD) include 2 women who had PGT but had no embryos below the threshold of heteroplasmy that was likely to cause disease and were subsequently referred for mitochondrial donation. Since the start of the mitochondrial reproductive care pathway, mitochondrial donation has resulted in 8 live births and 1 ongoing pregnancy, and PGT has resulted in 13 live births. "Complete" refers to completion of a full course of treatment. A total of 51 women who were not able to proceed with either PGT or mitochondrial donation, within their reproductive capacity, can be referred back to the MRA-C. BMI denotes body-mass index, ECG electrocardiography, HFEA Human Fertilisation and Embryology Authority, and NK not known.

left ventricle. This finding prompted a detailed pediatric cardiology assessment that showed ventricular preexcitation (Wolff–Parkinson–White pattern) and atrial tachycardia with secondary left ventricular dilatation, although the patient was asymptomatic. Extensive genetic investigations, including genome sequencing, did not identify a hereditary cause for this cardiac phenotype. Two independent pediatric cardiologists concluded that the atrial tachyarrhythmia was responsible for the enlarged left ventricle with reduced contractility. Treatment with antiarrhythmic medications has resulted in the left ventricle returning to normal morphologic features and function. This child was assessed at an adjusted age of 18 months 12 days, with the use of the Bayley Scales of Infant and Toddler Development, 3rd edition[16]: the scores were within the normal range. Both hearing and vision were assessed as normal by bedside testing. At the time of this report, the child remains well and continues to develop normally. Antiarrhythmic treatment is being slowly weaned with a view to possible ablative therapy in the future. One other child has had an 18-month neurodevelopmental assessment that was normal in all the domains. Planned cardiology assessments have been normal.

DISCUSSION

Women at risk for transmitting severe mtDNA disease to their children should have the opportunity to make informed choices about their reproductive options: prenatal testing, PGT, mitochondrial donation, egg donation, adoption, and deciding not to have children. Multidisciplinary advice regarding the optimal choice for an individual woman is tailored to the specific mtDNA variant and the woman's own views on risk reduction. In this study, many of the women who were eligible for mitochondrial donation or PGT decided, after counseling, not to proceed in the pathway at this time. Because mitochondrial donation and PGT involve in vitro fertilization, the success of which is inversely correlated with age,[18,19] we would encourage this type of early fact-finding discussion. Although this study involved women from the United Kingdom, we believe that the results are generalizable to other countries.

There is a theoretical risk of deterioration in health during pregnancy among women with pathogenic mtDNA variants, with an increased risk of pregnancy complications and early delivery.[20] In seven pregnancies, including one that is ongoing, no complications have been reported, although one patient with hypertrophic cardiomyopathy had intermittent atrial fibrillation. This patient also had a clinically significant complication of severe hyperlipidemia during pregnancy.[21] Despite extensive metabolic and genetic investigations (see the Results section in the Supplementary Appendix), the cause of this hyperlipidemia has not been identified. We are not aware of reports of extreme hyperlipidemia associated with this or any other mtDNA variant during pregnancy.

The eight infants were healthy at birth and are developing normally. Our data support a marked reduction in the transmission of pathogenic mtDNA variants after mitochondrial donation, with heteroplasmy either undetectable or well below a threshold likely to cause disease. We were able to sample only easily accessible tissues and do not know whether these tissues are entirely representative.

One child born to a mother with m.4300A→G variant–related disease had hyperlipidemia (also noted in her mother during pregnancy) and cardiac arrhythmia; both conditions have been successfully treated. Primary cardiac arrhythmia is

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Table 1. Characteristics of Women with Pregnancies after Pronuclear Transfer.*

mtDNA Variant (NC_012920.1)	Heteroplasmy				Age	Disease in Family	Outcome of Mitochondrial Donation
	Blood	Urine	Buccal	Muscle			
	percent				*yr*		
m.3260A→G	36	79	NA	94	27	Severe cardiomyopathy	Live birth
m.3460G→A	67	89	86	NA	31	Blindness	Live birth
m.4300A→G	Hom	Hom	Hom	NA	36	Fatal cardiomyopathy or child death	Live birth
m.4300A→G	Hom	Hom	NA	NA	34	Fatal cardiomyopathy or child death	Live birth
m.11778G→A	74	82	NA	81	35	Blindness	Live birth
m.11778G→A	Hom	Hom	Hom	NA	30	Blindness	Live births (twins)
m.11778G→A	94	Hom	Hom	NA	31	Blindness	Live birth
m.11778G→A	Hom	Hom	Hom	NA	33	Blindness	Ongoing pregnancy

* Hom denotes homoplasmic within the limits of detection (>97%), and NA not assessed.

Table 2. Details of Pregnancy and Births of Infants after Pronuclear Transfer.*

mtDNA Variant (NC_012920.1)	Max. Mat. Het.†	Sex	Antenatal Ultrasound	Gestation	Mode of Delivery	Birth Weight	Apgar Score‡			Neonatal Heteroplasmy	
							1 min	5 min	10 min	Blood	Urine
	%					*g*				%	
m.3260A→G	94	M	Normal	38 wk	EL LSCS	3220	—	9	10	16	20
m.3460G→A	89	F	Normal	38 wk 6 days	NVD	3090	—	9	10	ND	ND
m.4300A→G	>97	F	Abnormal§	38 wk	EL LSCS	2815	10	10	10	5	9
m.4300A→G	>97	M	Normal	38 wk 6 days	EL LSCS	3210	9	10	10	ND	ND
m.11778G→A	82	F	Normal	42 wk 2 days	EL LSCS	3665	5	7	9	ND	ND
m.11778G→A¶	>97	M	Normal	36 wk 1 day	EL LSCS	2460	5	9	10	ND	NA
m.11778G→A¶	>97	M	Normal	36 wk 1 day	EL LSCS	2190	5	9	10	ND	NA
m.11778G→A	>97	F	Normal	36 wk 2 days	NVD	3330	5	9	9	12	13

* EL LSCS denotes elective lower-segment cesarean section, NA not available, ND not detected, and NVD normal vaginal delivery.
† Shown is the maximum maternal variant heteroplasmy level detected in any sample. A value of more than 97% indicates homoplasmic within the limits of detection.
‡ Scores range from 0 to 10, with higher scores indicating a better physical condition of the newborn.
§ A suspected renal-tract abnormality was not detectable on a postpartum scan.
¶ Shown are details for a set of twins.

not a known feature of the m.4300A→G variant (rather, patients with this variant are at risk for progressive hypertrophic cardiomyopathy).[5] None of the other children had cardiac problems. We have instigated a pediatric cardiology review at 3 to 6 months in all the children born after mitochondrial donation. Another child had myoclonic epilepsy of infancy, a self-limiting condition that has resolved.

Although concerns exist about the safety of mitochondrial donation, including mismatch between nuclear DNA and mtDNA and reversion to the original maternal mitochondrial genotype,[22,23] it is challenging to establish cause and effect of adverse health outcomes in babies born. First, the procedures of conventional assisted reproductive technology are associated with an increased incidence of congenital anomalies, most notably of the cardiovascular system.[24] Second, some pathogenic mtDNA variants are associated with an increased

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incidence of pregnancy complications that can cause adverse health outcomes in children.[20] Third, the risk of cardiovascular defects is increased by exposure during early pregnancy to maternal metabolic conditions such as diabetes[25] and hyperlipidemia.[26] It is of interest that one child born after PGT, performed for m.3243A→G (a pathogenic mtDNA variant strongly associated with the development of diabetes mellitus) had a congenital cardiac defect; the maternal glycated hemoglobin level was normal before conception.

We found that pronuclear transfer, a form of mitochondrial donation, was effective in reducing the level of pathogenic mtDNA variant to substantially below the threshold for clinical disease in the offspring of women with homoplasmic (or high heteroplasmic) levels. We are assessing, over the long term, the health and extent of heteroplasmy (if detectable) of the offspring. Indeed, the role of mitochondrial donation as a choice for women with a heritable pathogenic mtDNA variant will only be established with the availability of additional data.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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REFERENCES

1. Gorman GS, Schaefer AM, Ng Y, et al. Prevalence of nuclear and mitochondrial DNA mutations related to adult mitochondrial disease. Ann Neurol 2015;77:753-9.
2. Mancuso M. Complex neurological and multisystem presentations in mitochondrial disease. Handb Clin Neurol 2023;194:117-24.
3. Stewart JB, Chinnery PF. Extreme heterogeneity of human mitochondrial DNA from organelles to populations. Nat Rev Genet 2021;22:106-18.
4. Carelli V, La Morgia C, Yu-Wai-Man P. Mitochondrial optic neuropathies. Handb Clin Neurol 2023;194:23-42.
5. Taylor RW, Giordano C, Davidson MM, et al. A homoplasmic mitochondrial transfer ribonucleic acid mutation as a cause of maternally inherited hypertrophic cardiomyopathy. J Am Coll Cardiol 2003;41:1786-96.
6. Ng YS, Bindoff LA, Gorman GS, et al. Mitochondrial disease in adults: recent advances and future promise. Lancet Neurol 2021;20:573-84.
7. Floros VI, Pyle A, Dietmann S, et al. Segregation of mitochondrial DNA heteroplasmy through a developmental genetic bottleneck in human embryos. Nat Cell Biol 2018;20:144-51.
8. Árnadóttir ER, Moore KHS, Guðmundsdóttir VB, et al. The rate and nature of mitochondrial DNA mutations in human pedigrees. Cell 2024;187(15):3904-3918.e8.
9. Steffann J, Frydman N, Gigarel N, et al. Analysis of mtDNA variant segregation during early human embryonic development: a tool for successful NARP preimplantation diagnosis. J Med Genet 2006;43:244-7.
10. Sallevelt SCEH, Dreesen JCFM, Drüsedau M, et al. Preimplantation genetic diagnosis in mitochondrial DNA disorders: challenge and success. J Med Genet 2013;50:125-32.
11. Greenfield A, Braude P, Flinter F, Lovell-Badge R, Ogilvie C, Perry ACF. Assisted reproductive technologies to prevent human mitochondrial disease transmission. Nat Biotechnol 2017;35:1059-68.
12. Hyslop LA, Blakeley P, Craven L, et al. Towards clinical application of pronuclear transfer to prevent mitochondrial DNA disease. Nature 2016;534:383-6.
13. Gorman GS, McFarland R, Stewart J, Feeney C, Turnbull DM. Mitochondrial donation: from test tube to clinic. Lancet 2018;392:1191-2.
14. Hyslop LA, Blakely EL, Aushev M, et al. Mitochondrial donation and preimplantation genetic testing for mtDNA disease. N Engl J Med. DOI: 10.1056/NEJMoa2415539.
15. Code of practice. 9th ed. London: Human Fertilisation and Embryology Authority, October 2023 (https://portal.hfea.gov.uk/media/yrkn55xa/2024-10-01-hfea-code-of-practice-v9-4.pdf).
16. Bayley N. Bayley scales of infant and toddler development. 3rd ed. San Antonio, TX: Pearson PsychCorp, 2006.
17. Zuberi SM, Wirrell E, Yozawitz E, et al. ILAE classification and definition of epilepsy syndromes with onset in neonates and infants: position statement by the ILAE Task Force on Nosology and Definitions. Epilepsia 2022;63:1349-97.
18. Shingshetty L, Cameron NJ, Mclernon DJ, Bhattacharya S. Predictors of success after in vitro fertilization. Fertil Steril 2024;121:742-51.
19. Wang P, Zhao C, Xu W, Jin X, Zhang S, Zhu H. The association between the number of oocytes retrieved and cumulative live birth rate in different female age strata. Sci Rep 2023;13:14516.
20. Hui L, Hayman P, Buckland A, et al. Pregnancy in women with mitochondrial disease — a literature review and suggested guidance for preconception and preg-

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nancy care. Aust N Z J Obstet Gynaecol 2025;65:30-6.
21. Goldberg AS, Hegele RA. Severe hypertriglyceridemia in pregnancy. J Clin Endocrinol Metab 2012;97:2589-96.
22. Fetterman JL, Zelickson BR, Johnson LW, et al. Mitochondrial genetic background modulates bioenergetics and susceptibility to acute cardiac volume overload. Biochem J 2013;455:157-67.
23. Kang E, Wu J, Gutierrez NM, et al. Mitochondrial replacement in human oocytes carrying pathogenic mitochondrial DNA mutations. Nature 2016;540:270-5.
24. Sargisian N, Petzold M, Furenäs E, et al. Congenital heart defects in children born after assisted reproductive technology: a CoNARTaS study. Eur Heart J 2024;45:4840-58.
25. Øyen N, Diaz LJ, Leirgul E, et al. Prepregnancy diabetes and offspring risk of congenital heart disease: a nationwide cohort study. Circulation 2016;133:2243-53.
26. Cao L, Du Y, Zhang M, et al. High maternal blood lipid levels during early pregnancy are associated with increased risk of congenital heart disease in offspring. Acta Obstet Gynecol Scand 2021;100:1806-13.
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